AH Statistics Essential Skills Booklet

Revise Your Notes → **Practise Essential Skills** → Work through SQA Papers

The purpose of this revision booklet

The best way to prepare for the AH Statistics exam is to **work through SQA exam papers**. But before you do this, it is important to **revise all the key skills** you have met across the Advanced Higher Statistics course. This booklet aims to give you a chance to test your understanding of those key skills, so that you know when you are ready to begin working through past exam papers. You should use your notes and examples from class to help structure your solutions, asking for help with anything you don't understand.

The worked solutions on Teams should also serve as a reminder of how to structure your solutions.

Chapter	Essential Skills
Probability Theory (Ch2, Ch4)	Formulae, key vocabulary, tree diagrams, contingency tables
Exploratory Data Analysis (Ch1)	Reading information from graphs and tables, calculating fences
Sampling Theory (Ch3)	Knowledge of the six sampling techniques, describing processes
Linear Regression I (Ch5)	Calculating correlation coefficients, regression lines, predictions
Random Variables (Ch6)	Tabulating probability distributions, finding $E(X)$ and $V(X)$, laws
Discrete Distributions (Ch7)	Discrete Uniform $U(k)$, Binomial $B(n,p)$, Poisson $Po(\lambda)$ distributions
Continuous Distributions (Ch8)	Continuous Uniform $U(a,b,)$, Normal Distribution $N(\mu,\sigma^2)$
The Sample Mean (Ch9)	Distribution of the sample mean, Central Limit Theorem
Hypothesis Tests (Ch10)	One-sample tests: z-test, t-test, z-test for proportion
Confidence Intervals (Ch11)	A z CI for μ, A t CI for μ, A z CI for population proportion
Chi-Squared Tests (Ch13, Ch14)	χ^2 goodness of fit tests, χ^2 tests for independence/association
2-Sample Parametric Tests (Ch15)	Two-sample tests: z-test, t-test, paired t-test, z-test for proportions
Non-Parametric Tests (Ch16, Ch17)	Wilcoxon tests, Paired Wilcoxon, Mann-Whitney
Linear Regression II (Ch18)	Residuals, R^2, Prediction vs Confidence Intervals, y on x vs x on y
Control Charts (Ch12)	Control charts for sample mean and proportion, out-of-control

Probability Theory

1. A camera store sells two models of Fuji camera: the XT4 and XT5. Each model is available in either a Black or Silver finish. 70% of the XT4s sold are in a Black finish, whilst 40% of the XT5s sold are in a Black finish.

   35% of the Fuji cameras sold by the store are XT4s.
   Calculate the probability that a Fuji camera the store sells in a Silver finish is an XT5.

2. Given exhaustive events F and G such that $P(F) = 0.3$ and $P(\bar{G}) = 0.1$, find $P(F|G)$.

3. A French wine merchant sells many different wines. The wines available are shown in the contingency table below, broken down by type and whether they are French or not.

	French	**Not French**
Red	24	18
White	30	12
Rosé	21	15

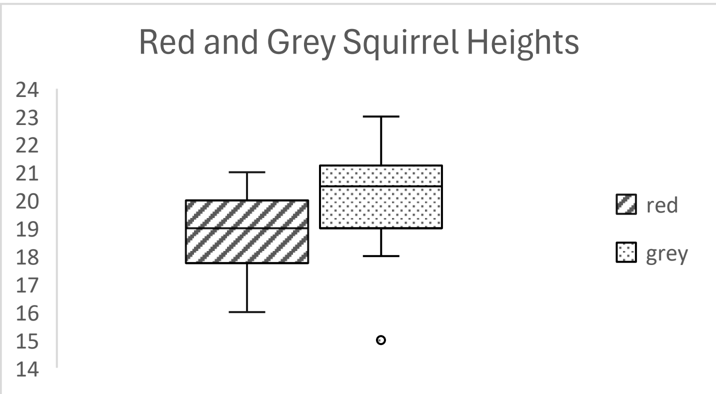
   A wine is selected at random.
   Let F be the event that the wine chosen is French, R be that the wine is red.
   Find $P(R|\bar{F})$.

Exploratory Data Analysis

4. The heights (in cm) of a sample of red squirrels and a sample of grey squirrels are recorded. The computer output below gives a selection of summary data for each sample.

data	n	mean	sd	Q1	Q3
red	27	18.7	1.68	17	20
grey	30	20.3	1.73	19	21.25

   The data from the samples is also represented below using boxplots.



   a) With reference to both the summary data **and** the boxplots, compare the average heights of the two types of squirrel.

   b) Show that the recorded height of 15 centimetres in the sample of grey squirrels is a possible outlier.

Sampling Theory

5. Describe the possible steps required to obtain a systematic sample of 5% of a population containing 240 members.

6. Given that a population can be described as belonging to three distinct groups, A, B and C, each with their own characteristics and of size 120, 150 and 90 respectively, describe the possible steps required to obtain a stratified sample of size 48.

7. A company that runs a chain of fast food restaurants has identified 60 of its suburban outlets that are each considered to serve a similar diverse range of profiles of customer. It wishes to survey, in person, a random sample of 120 customers. Give an example of how a two-stage cluster sample may be obtained, and explain why this type of sample may be desirable for the company.

8. Aside from those sampling techniques used in questions 5 to 7, list one further random sampling technique and two non-random sampling techniques.

Linear Regression I

9. 11 students are randomly selected to take part in a trial. They are asked to use an app to record the amount of sleep they get on a particular night, in hours, and then the following morning they each sit a test assessing their ability to use logic to quickly and accurately solve problems, giving them a score out of 100 for their performance. The results are:

Hours of sleep (x)	Score (y)
7	58
8.5	73
5	48
6	51
6.5	55
7	53
8	74
7.5	80
3	28
7	59
8	87

Summary statistics are:

$\Sigma x = 73.5$ $\quad \Sigma y = 666$ $\quad \Sigma xy = 4686$ $\quad \Sigma x^2 = 515.75$ $\quad \Sigma y^2 = 43142$

a) Calculate the correlation coefficient and interpret it.

b) Determine the equation of the least squares y on x regression line for this data.

c) Obtain predictions for the scores for students who get 5.5 hours and 10 hours sleep respectively, and comment on the reliability of these predictions.

Random Variables

10. The probability distribution for discrete random variable X is tabulated below.

x	2	3	4	5
$P(X = x)$	0.4	c	0.1	0.2

Calculate:

a) The value of c

b) $P(X > 3)$

c) $E(X)$

d) $V(X)$

e) $E(3X - 1)$

f) $V(3X - 1)$

11. Given independent random variables X and Y with means of 5 and 6 respectively, and standard deviations of 3 and 2 respectively, calculate $E(2X - Y)$ and $V(2X - Y)$.

Discrete Distributions

12. Given $X \sim U(7)$, calculate:

a) $P(X < 5)$

b) $P(2 \leq X \leq 5)$

c) $E(X)$

d) $V(X)$

e) $E(1 - 2X)$

f) $V(1 - 2X)$

13. Given $X \sim B(12, 0.4)$, calculate:

a) $P(X < 5)$

b) $P(X \geq 8)$

c) $P(4 < X < 7)$

d) $E(X)$

e) $V(X)$

f) $SD\left(\frac{1}{2}X\right)$

14. Given $X \sim Po(7.5)$, calculate:

a) $P(X > 8)$

b) $P(X \leq 4)$

c) $P(9 \leq X \leq 12)$

d) $E(X)$ and $V(X)$

e) $P\big(X < E(X)\big)$

f) $V(3 - 5X)$

Continuous Distributions

15. Given $X \sim U(2,7)$, calculate:

a) $P(X < 5)$

b) $P(2 \leq X \leq 5)$

c) $E(X)$

d) $V(X)$

e) $E\left(\frac{1}{10}X + 1\right)$

f) $V\left(\frac{1}{10}X + 1\right)$

16. The length of time a gardener takes to repot a species of plant is normally distributed with a mean of 110 seconds and a standard deviation of 15 seconds. Calculate:

a) The probability that the gardener takes longer than two minutes to repot a plant.

b) The probability that the gardener takes between 105 and 115 seconds to repot a plant.

c) The time t such that the gardener repots 95% of plants within t seconds.

17. The time it takes for a chef to chop an onion is normally distributed, with a mean of 45 seconds and a standard deviation of 10 seconds. Stating an assumption required, calculate the probability that the chef chops three onions in under two minutes.

18. Given $X \sim Po(40)$, use a suitable approximation, with justification, to calculate $P(X > 50)$.

19. Given $X \sim B(80,0.25)$, use a suitable approximation, with justification, to calculate $P(X < 15)$.

The Sample Mean

20. The mass of a red squirrel is normally distributed with a mean of 250 grams and a standard deviation of 8 grams. A conservationist studying red squirrels captures and weighs a sample of 12 squirrels. Calculate the probability that the sample mean obtained is greater than 252 grams.

21. The mean time it takes for a FI pupil to complete a cross country run is 32 minutes, and the standard deviation is 6 minutes. State, with justification, the distribution of the sample mean for a sample of 40 FI runners, and calculate the probability that the sample mean obtained is within one minute either side of the mean time.

Hypothesis Tests

22. It is stated that the mean height of squirrels living in some woodland is 20cm, and it is well established from previous observation that the heights of squirrels is normally distributed and the standard deviation for these squirrels is 3cm. A random sample of 10 squirrels gives a sample mean of 19cm. Perform a test to assess whether the heights of the squirrels has changed.

23. A sample is taken of 12 grey squirrels, finding a sample mean of 258 grams with a standard deviation of 15 grams. Perform a test to determine whether the mean height of a grey squirrel is different from 250 grams.

24. A wildlife blogger wishes to claim that, in March, over three-quarters of red squirrels still retain clearly visible ear tufts. Of a random sample of 35 red squirrels in March, 80% still have visible ear tufts. Perform a test to determine whether this data offers evidence to support the blogger's proposed claim.

Confidence Intervals

25. A random sample of 25 penalty kicks are observed, with the speeds of each kick measured. The mean kick speed observed is 45mph and the standard deviation is 9mph. Obtain a 95% confidence interval for the mean kick speed for a penalty.

26. The masses of a random sample of 12 burgers sold by a burger van are measured, with a sample mean of 205 grams obtains, and a standard deviation of 7 grams. Obtain a 95% confidence interval for the mean mass of the burgers sold by the van.

27. Out of a random sample of 140 houses in a town, 35 of them report having experienced problems with their internet connection. Obtain a 90% confidence interval for the proportion of houses in the town that have experienced problems with their internet connection.

Chi-Squared Tests

28. It is state that a mixed bag of tulip bulbs gives red, purple, yellow and pink tulips in the ratio 3:1:2:2. The contents of a large mixed bag of tulip bulbs provides the following:

Colour	Red	Purple	Yellow	Pink
Frequency	35	12	28	15

Perform a test to assess whether there is evidence to suggest that the tulip bulb colours are not distributed as stated.

29. Sixty pupils are randomly spotted, and their Form and whether they are wearing a tie is recorded. Assess the evidence for an association between Form and tie-wearing.

	Tie	No Tie
Form I	18	6
Form II	20	16

Two-Sample Parametric Tests

30. Samples of 25 rugby players and 24 football players are asked how far they run each week as part of their training. The sample of rugby players gives a mean of 12 km and a standard deviation of 2 km, whilst the sample of football players gives a mean of 14 km and a standard deviation of 3 km. Perform a test to assess whether there is evidence at the 1% level of significance to suggest that football players run further on average than rugby players in training.

31. A similar study took a sample of 12 tennis players and 15 cricketers. The sample of tennis players gave a sample mean of 15 km with a standard deviation of 4 km, and the sample of cricketers gave a sample mean of 9 km and a standard deviation of 2 km. Perform a test at the 10% level of significance to assess whether the data suggests that there is a difference in the distance run in training between tennis players and cricketers.

32. A sample of nine pupils have their scores, out of 100, in the first and second assessments of the Higher course collected. The results for the thirteen pupils are:

Pupil	1	2	3	4	5	6	7	8	9
First Assessment	56	65	53	32	51	80	80	65	98
Second Assessment	68	63	47	49	72	78	88	74	100

Stating an assumption required, perform a test at the 1% level of significance to determine whether there is evidence to suggest that pupils to better on the second assessment, on average.

33. In a city centre school, 34 out of a random sample of 50 pupils walk to school. In a rural school, 24 out of a random sample of 40 pupils walk to school. Assess whether the data gives evidence to support a claim that pupils in the city centre school are more likely to walk to school than pupils in the rural school.

Non-Parametric Tests

34. The finishing times for a random sample of nine students running 100 metres are recorded, in seconds:

14.1 12.3 15.3 11.7 12.8 13.0 12.4 16.1 12.9

Perform a non-parametric test to assess whether there is evidence to suggest that the median time taken for a student to run 100 metres is different than 13 seconds.

35. It is decided that the test from Q32 should be repeated, but a non-parametric test be performed instead. Perform this test on the data, again at the 1% level of significance.

36. Five randomly selected FIII pupils sit a Maths Challenge, and score 34, 45, 50, 61 and 75.
Six randomly selected FIV pupils sit the same challenge, and score 34, 44, 57, 74, 81 and 95.
Perform a non-parametric test to assess whether there is evidence at the 5% level that FIV will tend to do better in the Maths Challenge than FIII pupils.

Linear Regression II

37. 11 students are randomly selected to take part in a trial. They are asked to use an app to record the amount of sleep they get on a particular night, in hours, and then the following morning they each sit a test assessing their ability to use logic to quickly and accurately solve problems, giving them a score out of 100 for their performance. The results are:

Hours of sleep (x)	7	8.5	5	6	6.5	7	8	7.5	3	7	8
Score (y)	58	73	48	51	55	53	74	80	28	59	87

Summary statistics are:

$\Sigma x = 73.5$ $\quad\Sigma y = 666$ $\quad\Sigma xy = 4686$ $\quad\Sigma x^2 = 515.75$ $\quad\Sigma y^2 = 43142$

(You may wish to refer to your calculations from Q9 to complete these questions).

a) Calculate the coefficient of determination, and interpret it in context.

b) Obtain a 95% prediction interval for a student who obtains 5.5 hours of sleep.

c) Obtain a 95% confidence interval for a student who obtains 5.5 hours of sleep.

d) Obtain the equation of the least squares regression line for x on y.

Control Charts

38. In a ceramics factory, historically 8% of the bowls produced contain serious defects. Random samples of 60 bowls are checked daily and the number of defective bowls are counted. A *p*-chart is used to monitor the performance of the manufacturing. Determine the 1-sigma limits for this chart.

39. A machine dispenses olive oil into bottles. The volume of oil dispenses into a bottle is known to be normally distributed, with a mean of 502ml and a standard deviation of 3ml. Samples of 8 bottles are checked each hour, and the volume of oil in each measured, for which a *x*-bar control chart is produced. Determine the upper and lower control limits for this chart.

40. A simplified control chart is shown below. Determine whether the process is in control.